

SE LAB 3 – ARCHITECTURE SELECTION & JUSTIFICATION

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Problem Statement

Customizable Subscription Box Scheduler

Step 1: Scenario Review

Assigned Scenario

Customizable Subscription Box Scheduler

Key Functional Requirements

1. The system shall allow subscribers to select products for their monthly box based on preferences.
2. Subscribers shall be able to customize their monthly box.
3. Subscribers shall be able to swap box items.
4. Subscribers shall be able to pause or skip deliveries.
5. Subscribers shall be able to view their subscription status.
6. The system shall support subscription payment processing.
7. The fulfillment lead shall be able to generate a fulfillment manifest containing product and delivery details.

Key Non-Functional Requirements

1. Performance: The system should support exporting shipping labels for 10,000 monthly boxes in under 1 minute.
2. Security: Subscriber account and subscription information should require authenticated access.
3. Usability: The system should allow subscribers to easily select products, customize their box and view their subscription information.

Main Challenges

1. Performance

- The system needs to efficiently handle a large number of subscription boxes.
- Fulfillment information and shipping labels need to be generated quickly.

2. Usability

- Subscribers should be able to easily customize their boxes.
- Product selection, swapping and pause/skip options should be simple to use.

3. Security

- Subscriber and subscription information must be protected.
- Only authenticated users should be allowed to access account and subscription information.

Step 2: Architectural Style Analysis

1. Layered Architecture

Structure:

The system is divided into layers such as Presentation, Business and Data.

Advantages:

- Clear separation of concerns.
- Easy to understand and maintain.
- Suitable for traditional applications.

Disadvantages:

- Communication between layers can introduce performance overhead.
- Layers can become tightly coupled.
- Individual components are more difficult to scale independently.

Suitability for our system:

It provides a clear separation between subscriber interaction, subscription/business operations and data storage.

2. Microservices Architecture**Structure:**

The system is divided into small, independently deployable services.

Advantages:

- Individual services can be scaled independently.
- Provides fault isolation.
- Supports independent deployment.

Disadvantages:

- More operational complexity.
- Network communication can introduce latency.
- Data consistency can be difficult to manage.

Suitability for our system:

It could support independent scaling of payment and fulfillment services, but it may introduce unnecessary complexity for this system.

3. Client-Server Architecture**Structure:**

Multiple clients communicate with a centralized server.

Advantages:

- Centralized control.
- Simple deployment.
- Easier data consistency.

Disadvantages:

- Server can become a single point of failure.
- Scalability can become a bottleneck.

- Limited fault tolerance.

Suitability for our system:

It provides centralized management but may become a bottleneck when handling high-volume subscription and fulfillment operations.

Step 3: Architecture Selection & Component Identification

Selected Architecture

Layered Architecture

Reason for Selection:

1. It provides clear separation between the Presentation, Business and Data layers, making the subscription system easier to understand and maintain.
2. It allows different responsibilities such as subscription management, payment processing, fulfillment and data storage to be handled separate.

Component Identification

1. Order Manager Component

Responsible for:

- Managing subscriptions
- Customizing boxes
- Selecting products
- Swapping items
- Pausing/skipping deliveries
- Viewing subscription status

2. Payment Service Component

Responsible for:

- Processing subscription payments

- Handling payment requests
- Maintaining payment status

Remaining 3 Components

3. User Interface Component

Responsible for:

- Subscriber interaction
- Product selection
- Box customization
- Viewing subscription status
- Login/authentication

4. Fulfillment / Manifest Component

Responsible for:

- Generating fulfillment manifests
- Managing product and delivery details
- Supporting box preparation and delivery

5. Database Component

Responsible for storing:

- Subscriber information
- Product information
- Subscription details
- Payment information
- Delivery information

Final 5 Components

No.	Component	Main Responsibility
1	User Interface Component	Subscriber interaction
2	Order Manager Component	Subscription and box management
3	Payment Service Component	Payment processing
4	Fulfillment / Manifest Component	Fulfillment and delivery details
5	Database Component	Store system data

Component Diagram

